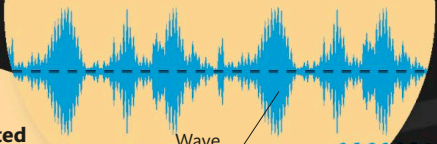
Digital sound

Digital sound is stored as large collections of binary numbers. The numbers describe the oscillations of an audio signal—an electrical copy of the sound wave of the original sound. Playing back a sound involves using electronic circuits that can reconstruct the audio signal from the numbers and play it through a loudspeaker.

Analogue to digital to analogue

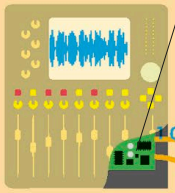
Digitization begins with an audio signal—an electrical copy, or analogue, of the sound wave. Typically, this comes from a microphone (see p.138). An analogue-to-digital converter measures the voltage of the audio signal thousands of times every second. It assigns each of the measurements, or samples, a number depending on the strength of the voltage. The numbers are stored in binary form (see p.158). To play back the sound, a sound signal must be produced and sent to a loudspeaker (see p.139) or headphones. This is done by a digital-to-analogue converter.

4 Signal processing
The sound now exists as a sequence of binary numbers. It can be processed with effects or filters and mixed with other sounds.

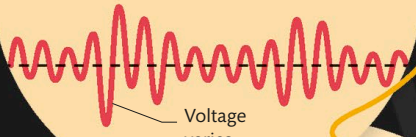


Wave formed of 1s and 0s

3 Signal converted
An analogue-to-digital converter (ADC) measures the voltage and assigns binary numbers to each sample.

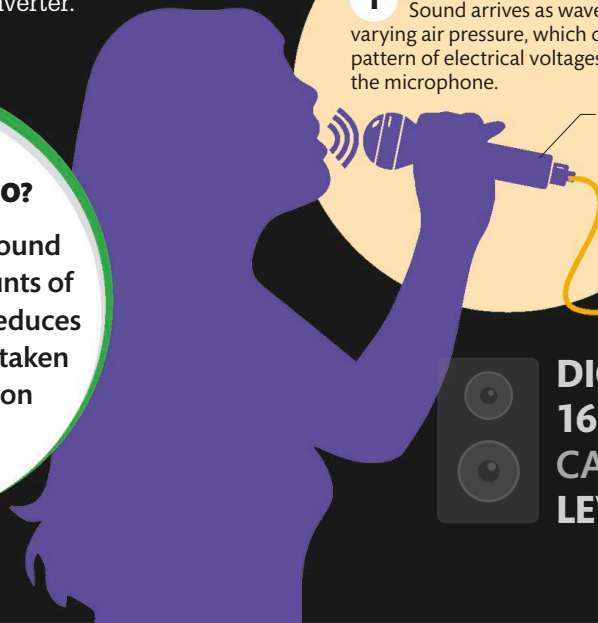


2 Cable carries signal
The varying voltage in the microphone cable is the audio signal—a copy, or analogue, of the rapidly varying air pressure.



Voltage varies

1 Sound captured
Sound arrives as waves of varying air pressure, which create a pattern of electrical voltages inside the microphone.



Microphone captures analogue audio signal

WHAT IS COMPRESSED AUDIO?

Good-quality digital sound can take up huge amounts of storage. Compression reduces the amount of storage taken up, with little effect on sound quality.

DIGITAL SOUND WITH 16 BITS PER SAMPLE CAN MEASURE 65,536 LEVELS OF VOLTAGE